

Visualizing AI: From Generative Models to Explainability

Visual Encodings · Representation Design · Model Interpretation · XAI

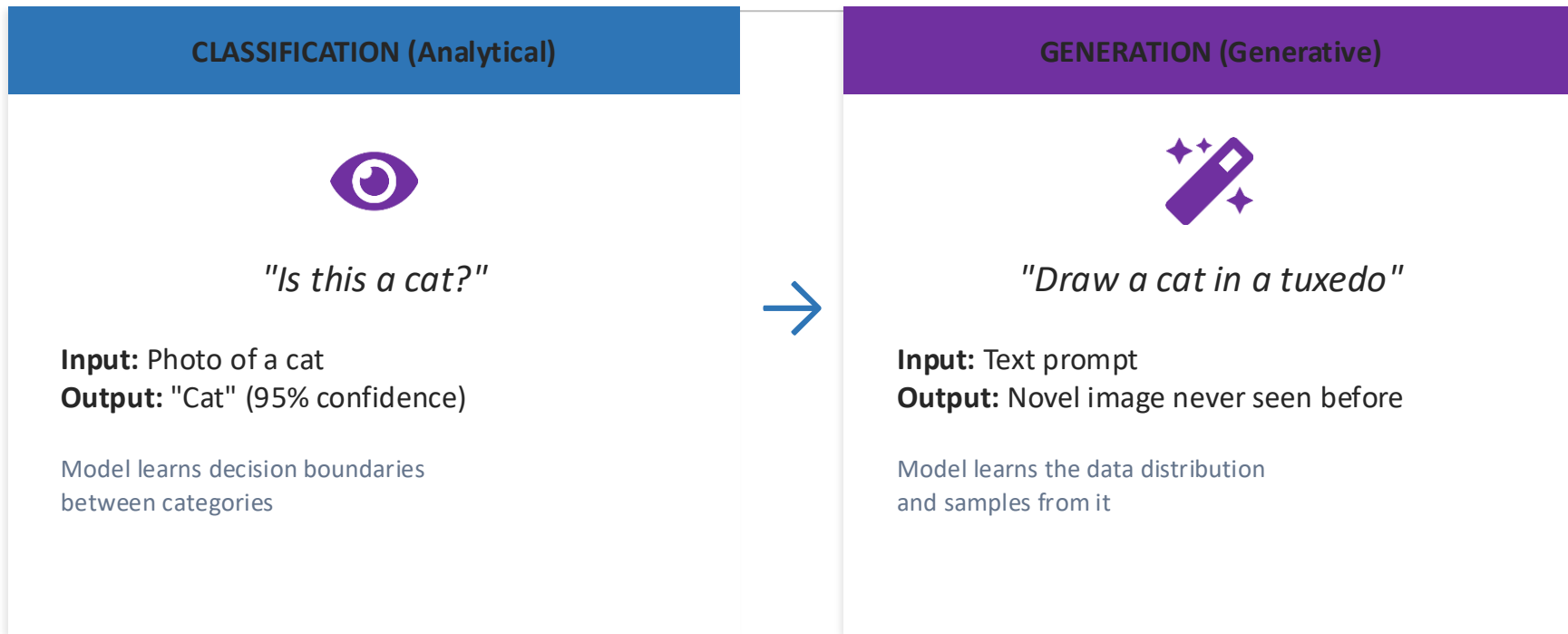
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Information Visualization

Visualizing Two Paradigms: Analytical vs. Generative

How visual representations differ when models classify vs. create.



Key Insight: Generative AI creates novel data samples that follow the distribution of the training set.

The Modality Matrix: Mapping Input/Output Spaces

A 2x3 visual taxonomy of generative AI capabilities.



Text → Text

ChatGPT, Claude, Gemini
Summarization, translation, code



Text → Image

Midjourney, DALL·E, Stable Diffusion
Photorealistic renders from prompts



Text → Video

Runway Gen-3, Sora, Kling
Full motion clips from text descriptions



Text → Audio

ElevenLabs, Suno, MusicGen
Voice cloning, music composition



Image → 3D

Point-E, TripoSR, Meshy
3D models from single photographs

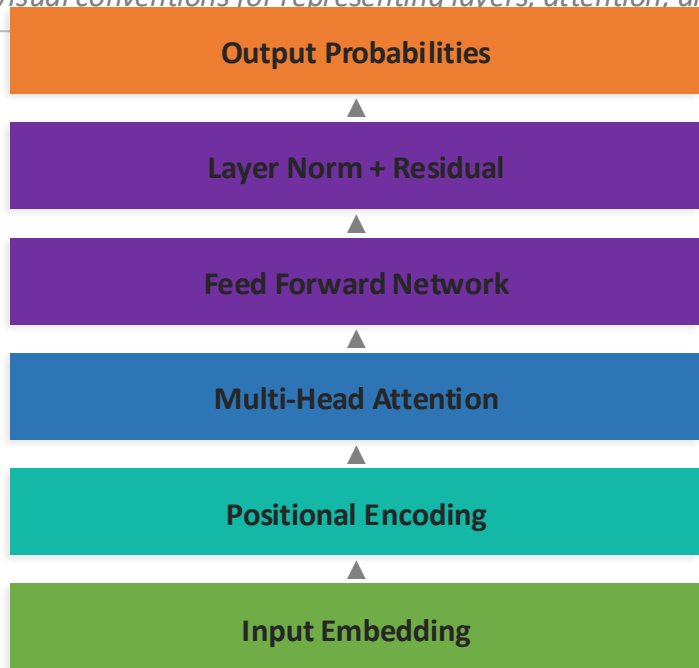


Text → Code

GitHub Copilot, Cursor, Claude
Full applications from descriptions

Diagramming the Transformer Architecture

Visual conventions for representing layers, attention, and data flow.



Self-Attention Mechanism

"The cat sat on the mat because it was tired"

Self-attention lets the model weigh the importance of every word relative to every other word simultaneously.

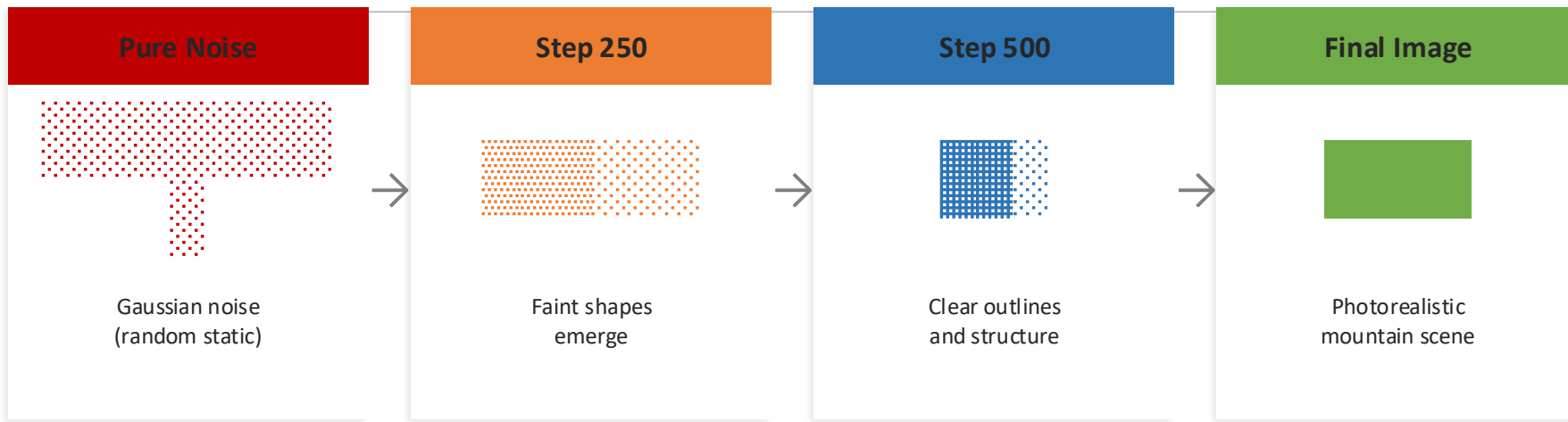
"it" attends strongly to "cat" → the model understands the pronoun reference.

"Attention Is All You Need" (2017)

Vaswani et al. — replaced RNNs with pure attention.
Enabled parallel training → GPT, BERT, LLaMA, Claude.

Visualizing Diffusion: A Step-by-Step Process Diagram

Sequential visual encoding of the forward and reverse denoising process.



How Diffusion Works

Forward Process: Gradually add Gaussian noise to an image until it becomes pure static.

Reverse Process: Train a neural network (U-Net) to predict and remove noise step-by-step.

Text conditioning: A text encoder (CLIP) guides which noise to remove, steering toward the prompt.

Visualizing Tokenization: Color-Coded Segmentation

Using color blocks and sequence diagrams to make subword splitting visible.

Example sentence broken into tokens:



Autoregressive Prediction

$P(w_n \mid w_1, w_2, \dots, w_{n-1})$ – "Given all previous tokens, what comes next?"

Subword Tokenization

Words are split into pieces.
"Visualization" → "Visual" + "ization".
This lets models handle any word,
even unseen ones.

Vocabulary Size

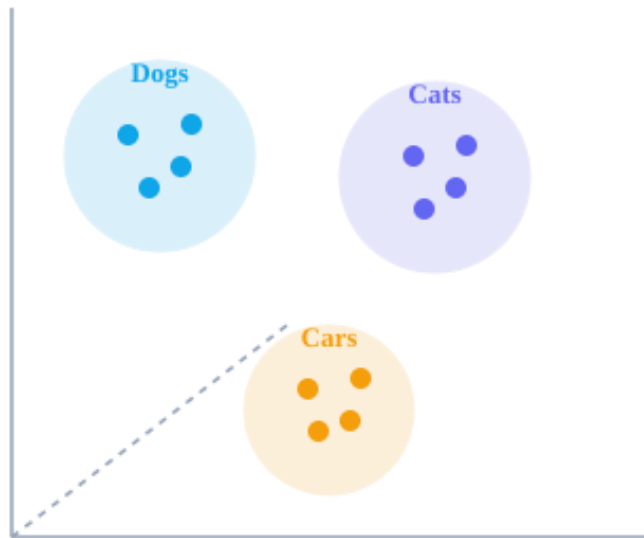
GPT-4: ~100K tokens. Claude: ~100K
tokens. Larger vocab = fewer tokens
per sentence = faster processing.

Context Window

The number of tokens a model can
process at once. GPT-4: 128K,
Claude: 200K. Longer = more
memory.

Projecting Latent Space: From 512D to 2D Scatter Plots

Dimensionality reduction as a visualization technique for embeddings.



High-Dimensional Latent Space (projected to 2D)

What Is Latent Space?

AI doesn't store raw images or text. It compresses them into coordinates in a high-dimensional mathematical space. Similar concepts cluster together.

Why It Matters

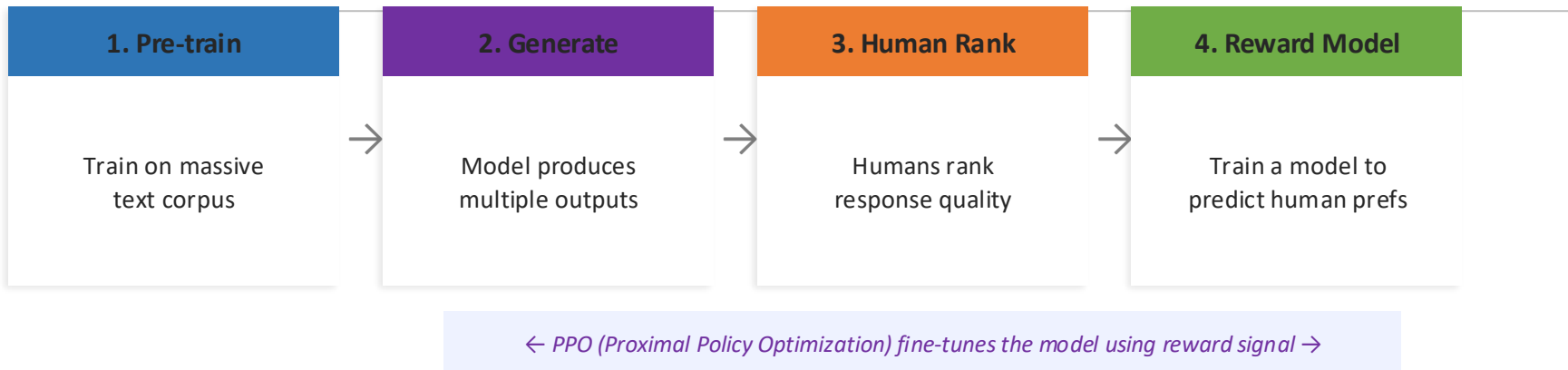
Interpolation: Move between points to blend concepts (e.g., morph a dog into a cat). Arithmetic: King – Man + Woman $\approx$ Queen.

Dimensionality

Real latent spaces have 512–4096 dimensions. We project them down to 2D/3D for visualization using T-SNE or UMAP.

Visualizing the RLHF Pipeline: Flow Diagrams & Loops

Using process diagrams to represent multi-stage training feedback loops.



Why AI Sounds Polite

Before RLHF, models produce raw completions — sometimes toxic, off-topic, or unhelpful. RLHF teaches the model to follow instructions, be helpful, and refuse harmful requests. This is why ChatGPT says "I'd be happy to help" instead of raw text.

DPO: The New Approach

Direct Preference Optimization (DPO) skips the reward model entirely. Instead of training a separate model, DPO directly optimizes the language model using human preference pairs. Simpler, cheaper, often equally effective.

Visual Case: Coca-Cola "Create Real Magic"

How generative outputs were curated, displayed, and evaluated at scale.

The Campaign

Coca-Cola partnered with OpenAI and Bain & Company in 2023 to let fans create AI artwork using brand assets with Stable Diffusion and GPT-4.

How It Worked

Users could remix iconic Coca-Cola imagery — polar bears, contour bottles, vintage ads — into original compositions via a web app.

Results

120,000+ submissions in 3 weeks. Selected artworks displayed on billboards in Times Square and Piccadilly Circus. Massive engagement boost.

Lesson: AI as a creative collaborator, not a replacement. Brand identity + user creativity + AI generation = viral engagement.

Visual Case: GitHub Copilot — Code as Overlay

Ghost text, inline suggestions, and stat dashboards as UI visualization.



```
def calculate_bmi(weight, height):  
    """Calculate Body Mass Index"""  
    bmi = weight / (height ** 2)  
    return round(bmi, 2)  
  
def classify_bmi(bmi):  
    if bmi < 18.5:  
        return "Underweight"  
    ← Copilot suggestion (gray = AI-generated)
```

55%

faster task completion

46%

of code written by AI

1.8M

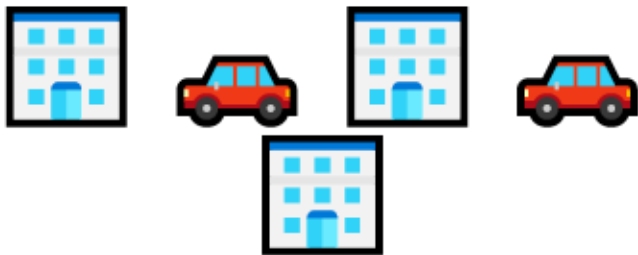
paying subscribers

Trained on public GitHub repos → LLM predicts next code tokens in real-time as you type.

Visual Case: AI-Generated Urban Renderings

Before/after comparison as a visual evaluation technique for AI outputs.

BEFORE: Standard Street



Concrete, cars, minimal green space
Pedestrian-hostile design

AFTER: AI-Redesigned



Green corridors, bike lanes, mixed-use
Human-centric design via AI rendering

Tools Used

Midjourney, LookX, Stable Diffusion XL
ControlNet for structural guidance

Impact

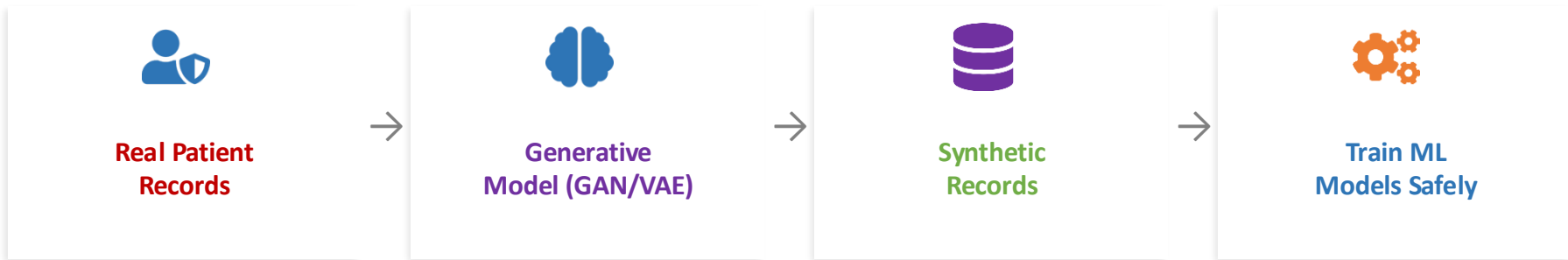
10x faster concept iteration. Clients see
photorealistic futures before breaking ground.

Visualizing Synthetic Data Pipelines

Flow diagrams for data generation, privacy, and downstream training.



The Problem: Medical data is locked behind privacy laws (HIPAA, GDPR).



What Makes It Work

Statistical fidelity: distributions match real data.
Privacy guarantees: no individual can be re-identified.
Downstream utility: ML models trained on synthetic data achieve 95%+ of real-data accuracy.

Key Players

Synthea: open-source synthetic patient generator.
Mostly AI: enterprise synthetic data platform.
Gretel.ai: privacy-preserving synthetic data APIs.
Google Health: synthetic EHR datasets.

Visualizing the Copyright Debate: Scales & Panels

Using visual metaphors (balance scales, side-by-side) for legal tensions.

FAIR USE ARGUMENTS

- Transformative use: AI creates new works
- Models learn patterns, not memorize
- Benefits society with new capabilities
- Similar to how humans learn from reading



INFRINGEMENT ARGUMENTS

- Training on copyrighted work without consent
- Can reproduce near-exact copies
- Displaces original creators economically
- No compensation to data contributors

NYT vs. OpenAI (2023–Present)

The New York Times sued OpenAI and Microsoft, alleging that GPT models were trained on millions of NYT articles without permission. The suit showed GPT-4 could reproduce near-verbatim NYT text. This is the landmark case that may define AI copyright law for a generation. Other suits: Getty Images vs. Stability AI, Authors Guild vs. OpenAI, and numerous artist class actions.

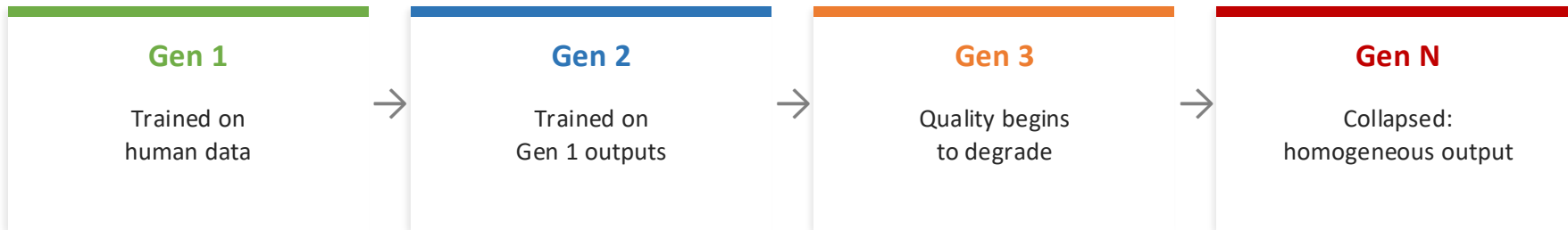
Visualizing Model Collapse: Generational Degradation

Sequential diagrams showing quality loss across AI training generations.



What happens when AI trains on AI-generated content?

As AI-generated content floods the internet, future models inevitably train on it. Research shows this creates a degenerative feedback loop where each generation produces lower-quality, less diverse outputs — a phenomenon called Model Collapse.



By 2026, an estimated 90%+ of internet content may be AI-generated. The "Dead Internet Theory" posits that most online interactions are already between bots.

Visualizing Multi-Modal Systems: Capability Matrices

Card-based visual taxonomy for mapping agent modalities and design.

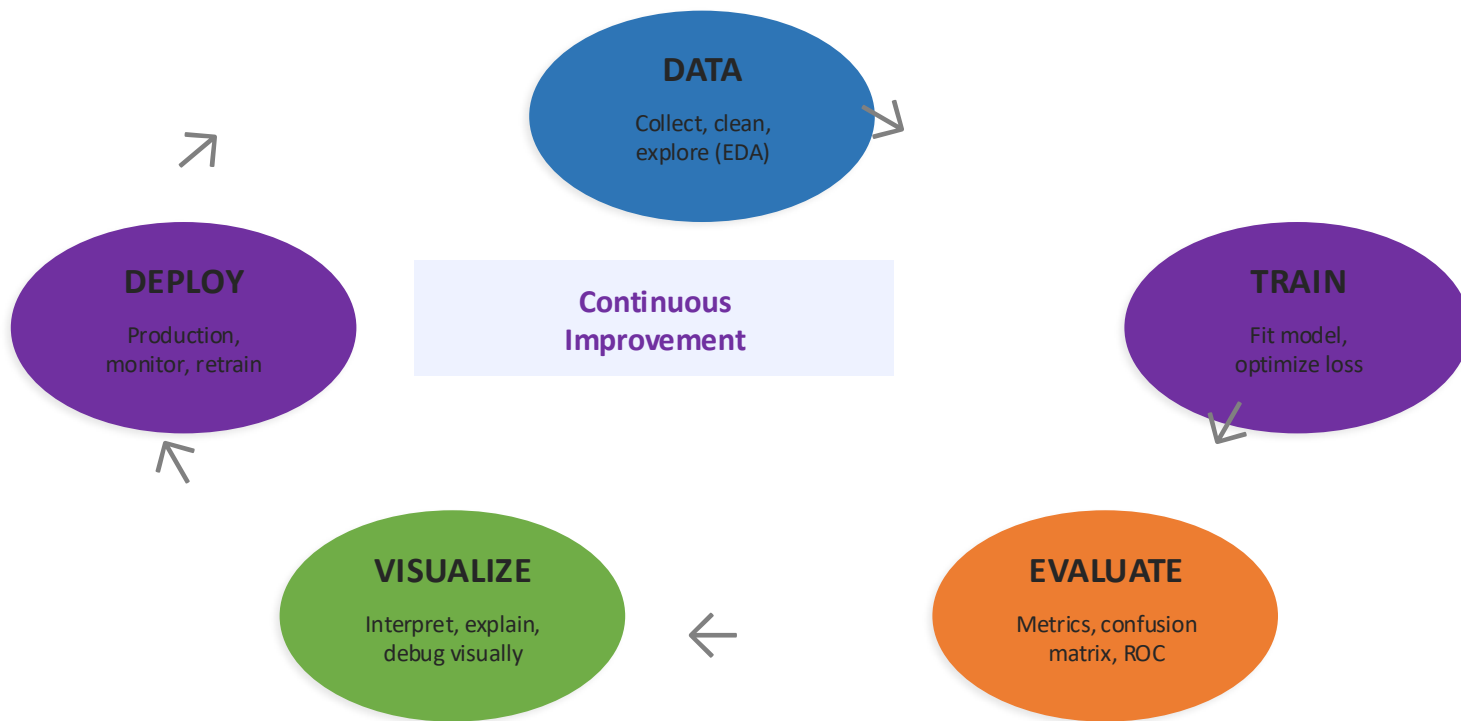
The evolution from reactive chatbots to proactive agents. Modern AI systems combine multiple modalities — vision, speech, language, and action — to autonomously plan multi-step tasks, use tools, browse the web, and execute real-world operations.

See	Hear	Write	Act
Process images, video, and screen content. Understand UI layouts and read documents.	Real-time speech transcription, speaker ID, and tone analysis via voice interfaces.	Generate text, code, and documents — the most mature output modality today.	Browse websites, fill forms, call APIs, execute code, and complete real tasks.

Emerging agents: Claude Computer Use · Devin (Cognition) · AutoGPT · Microsoft Copilot · Google Gemini Agents

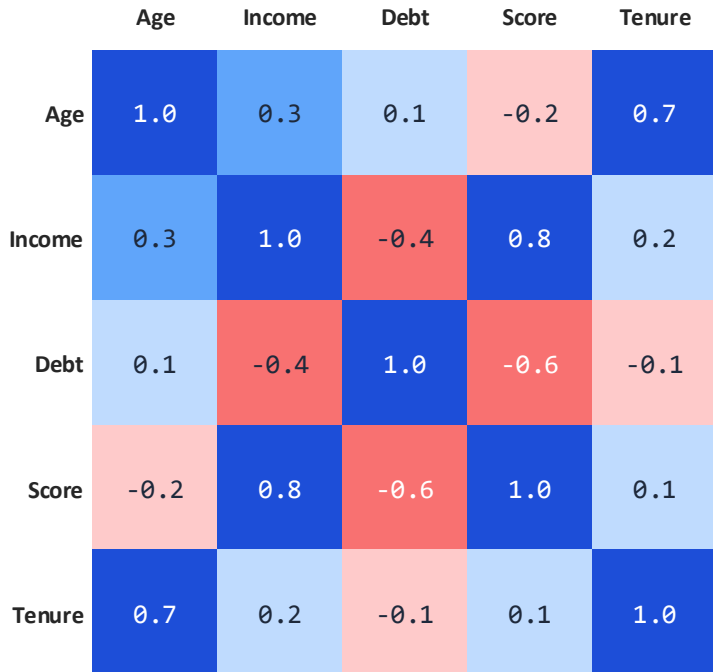
The ML Lifecycle as a Circular Flow Diagram

Why cycle diagrams outperform linear sequences for iterative processes.



Correlation Heatmaps: Color-Encoding Relationships

Design choices: diverging color scales, annotation, and when heatmaps fail.



Reading the Heatmap

Darker blue = stronger positive correlation.
Darker red = stronger negative correlation.
Diagonal is always 1.0 (self-correlation).

Income & Score = 0.8 → highly correlated.
Debt & Score = -0.6 → inversely related.

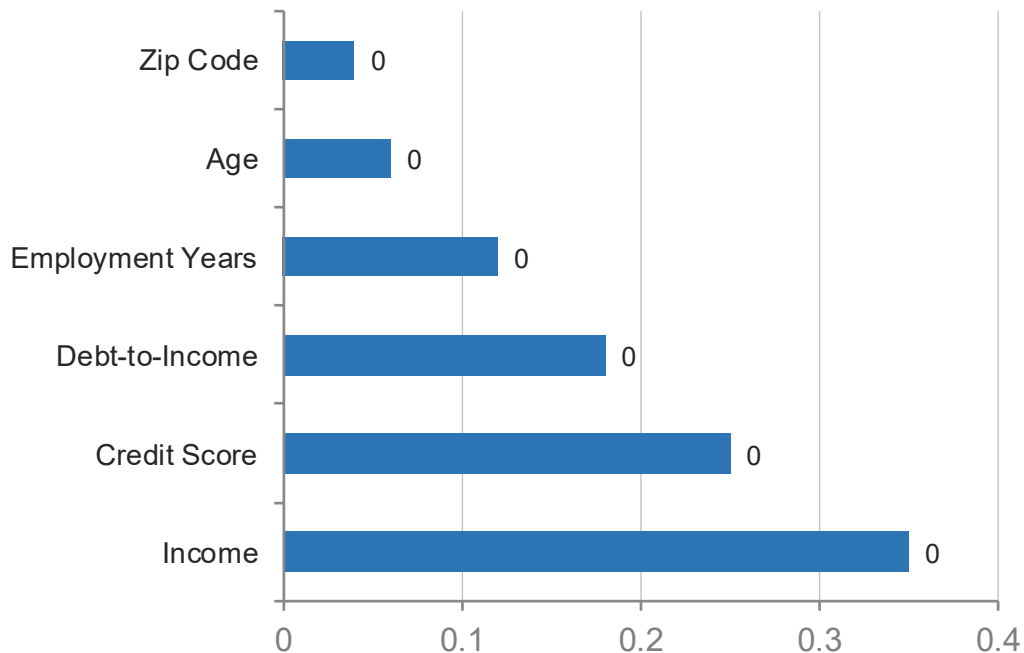
Teacher's Tip

"If two features are nearly identical (corr > 0.9), delete one to save compute and reduce multicollinearity. Use VIF (Variance Inflation Factor) for rigorous checking."

Tools: `seaborn.heatmap()`, `pandas.corr()`, `plotly.imshow()`

Horizontal Bar Charts for Feature Ranking

Why sorted horizontal bars outperform vertical bars and pie charts for rankings.



What This Tells Us

Income is the #1 predictor of loan approval — 35% of the model's decision power. Credit Score follows at 25%. Together, just two features drive 60% of predictions.

Methods

Permutation Importance: shuffle one feature, measure accuracy drop.
Gini Importance: built into tree models.
SHAP: game-theory attribution (see Slide 31).

The Confusion Matrix: A 2x2 Visual Encoding of Errors

Color-coded cells, derived metrics, and the most important ML visualization.

		PREDICTED	
		Positive	Negative
ACTUAL	Pos	TP 850	FP 50
	Neg	FN 30	TN 70

Reading the Matrix

TP = Correctly predicted positive. TN = Correctly predicted negative. FP = False alarm (Type I error). FN = Missed detection (Type II error).

Derived Metrics

Precision = $TP / (TP + FP) = 850 / 900 = 94.4\%$

Recall = $TP / (TP + FN) = 850 / 880 = 96.6\%$

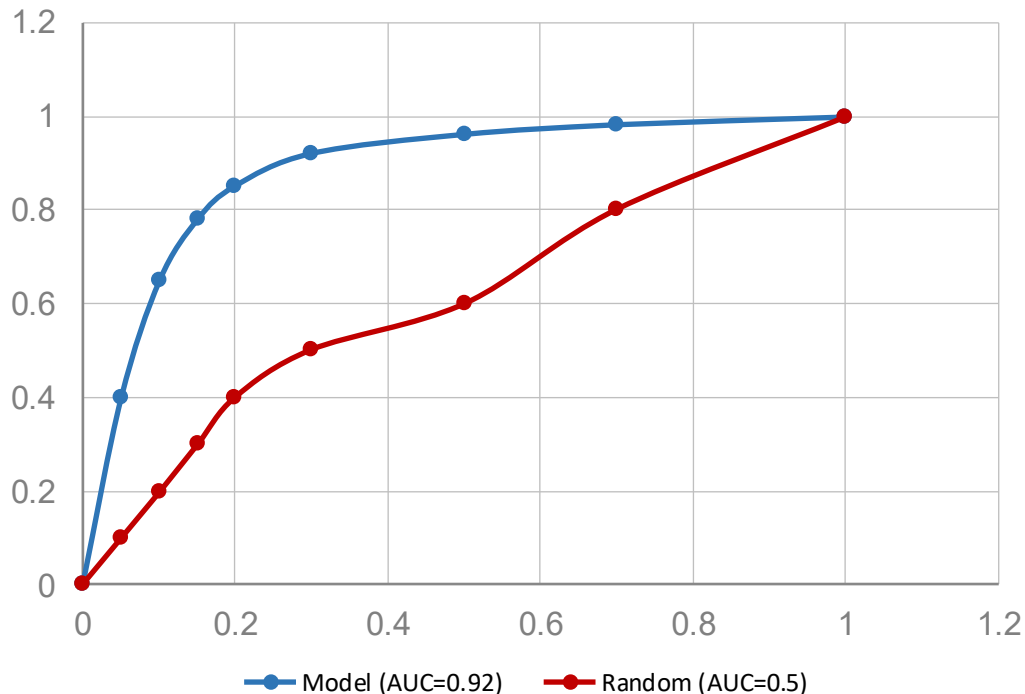
Accuracy = $(TP + TN) / \text{Total} = 920 / 1000 = 92\%$

The Trap of Accuracy

If 95% of emails are NOT spam, a model that always says "not spam" gets 95% accuracy but catches zero spam. Always check Precision & Recall!

ROC Curves: Line Charts for Threshold Trade-offs

Dual-axis plots, comparing curves, and AUC as a single-number summary.



What Is ROC?

Receiver Operating Characteristic plots TPR vs FPR at every classification threshold. The diagonal = random guessing. The further above the diagonal, the better.

AUC Score

Area Under the Curve. AUC=1.0: perfect. AUC=0.5: random. AUC<0.5: worse than random (flip predictions!).

When to Use

Best for binary classification with imbalanced classes. Compare models: higher AUC = better overall discriminative power.

Loss Curves: Dual-Line Time Series for Diagnostics

Gap analysis between train/validation lines as a visual overfitting detector.



What to Look For

Both curves dropping together = learning.
Gap growing = overfitting.
Both flat = underfitting (model too simple).

Overfitting

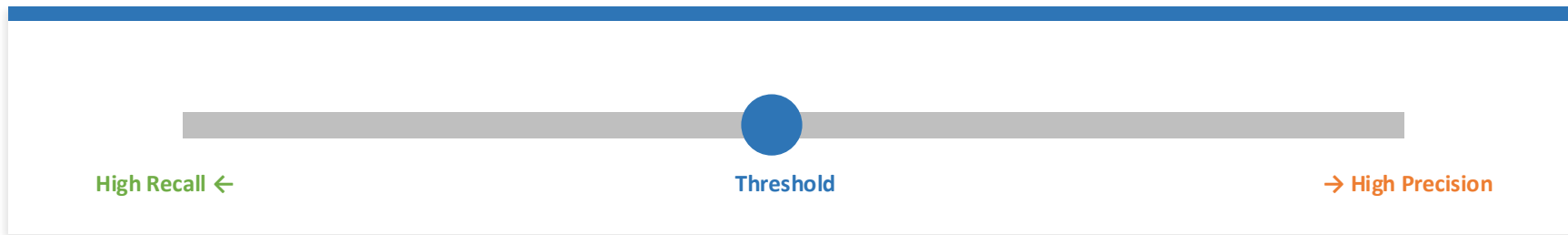
Training loss keeps dropping but validation loss rises. The model memorizes training data instead of learning patterns.

Solutions

Early stopping, dropout, data augmentation, regularization (L1/L2), reduce model complexity.

Visualizing Trade-offs: Sliders, Scenarios & Panels

Interactive slider metaphors and scenario panels for communicating trade-offs.



CANCER SCREENING

Priority: HIGH RECALL

We'd rather have false alarms (low precision) than miss a tumor (low recall). Missing cancer = patient could die. Cost of a false alarm = extra tests.

"Better safe than sorry"

SPAM FILTER

Priority: HIGH PRECISION

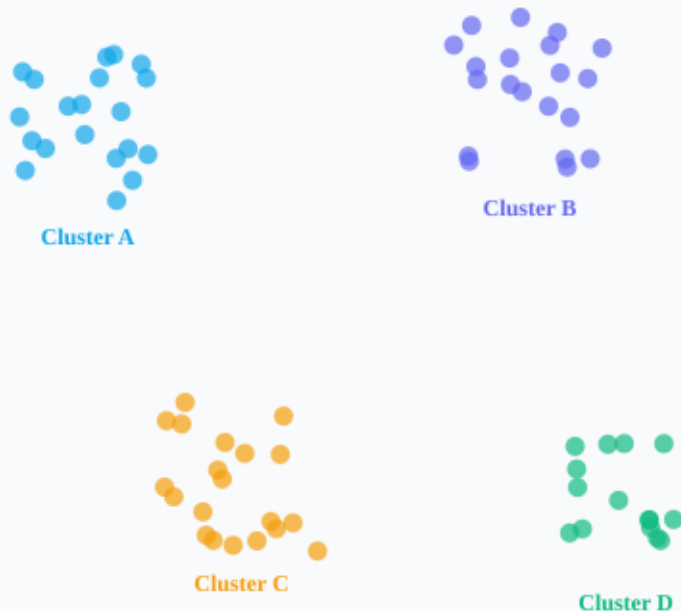
We'd rather let some spam through (low recall) than delete a real email (low precision). Missing spam = annoyance. Deleting real email = lost business.

"Don't throw out the baby with the bathwater"

T-SNE & UMAP: Projecting High Dimensions to 2D

Dimensionality reduction as a core visualization technique and its pitfalls.

2D Projection of High-Dimensional Data



T-SNE

t-Distributed Stochastic Neighbor Embedding. Preserves local structure: nearby points stay nearby. Great for visualization, but non-deterministic and slow on large datasets.

UMAP

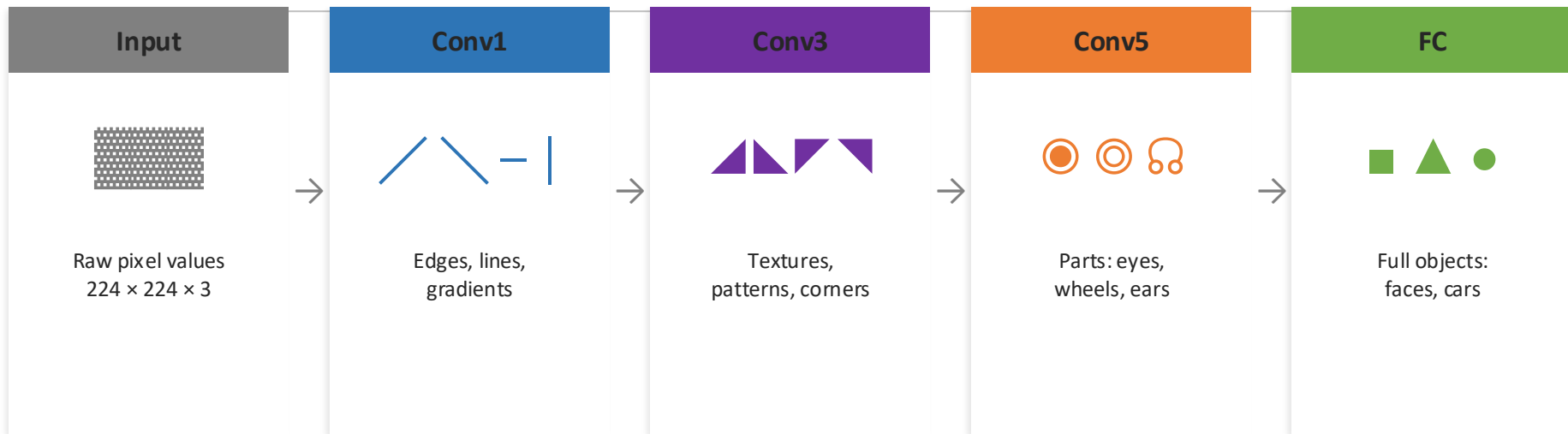
Uniform Manifold Approximation & Projection. Faster, preserves both local AND global structure. Deterministic with a fixed seed. Preferred for large-scale analysis.

Use Cases

Visualizing word embeddings, image clusters, customer segments, anomaly detection, and understanding what a neural network has learned.

Visualizing CNN Feature Hierarchies Layer-by-Layer

Sequential panel design showing how neural networks build representations.



The Hierarchy of Features

Early layers detect universal primitives (edges, gradients). Middle layers combine these into textures and patterns. Deep layers recognize complete objects. This hierarchy is remarkably consistent across architectures and datasets — it mirrors how the human visual cortex processes information.

Attention Maps: Weighted Connection Diagrams

Encoding attention weights via color intensity, line thickness, and arcs.

Visualizing Self-Attention



The model uses attention weights to resolve coreferences. When processing "it", the self-attention mechanism strongly attends to "cat" (weight 0.82), correctly identifying the antecedent.

Multi-Head Attention

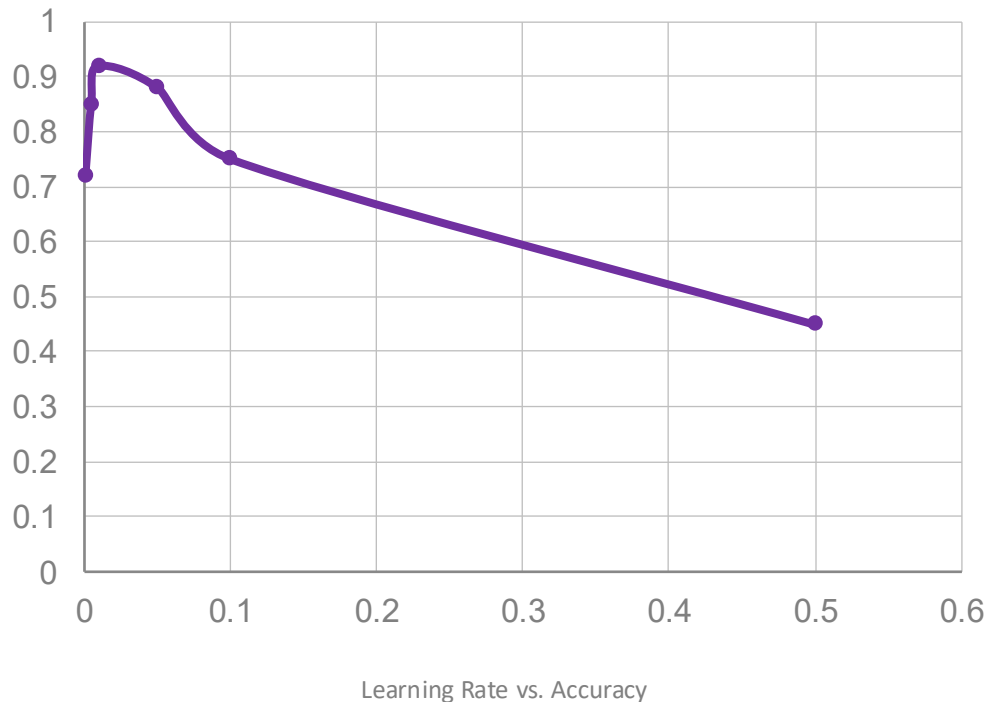
Models use 12-96 attention heads simultaneously. Each head learns different patterns: syntax, semantics, coreference, position. Visualizing all heads reveals the model's understanding.

Tools

BertViz: interactive attention visualizer.
Ecco: NLP model interpretation.
Transformer Debugger (OpenAI): token-level analysis of GPT models.

Surface Plots & Scatter for Hyperparameter Search

Choosing between 3D surfaces, 2D heatmaps, and parallel coordinates.



Surface Plots

3D surface plots show accuracy as a function of two hyperparameters. Peaks = optimal configurations. Valleys = poor performance regions.

Search Strategies

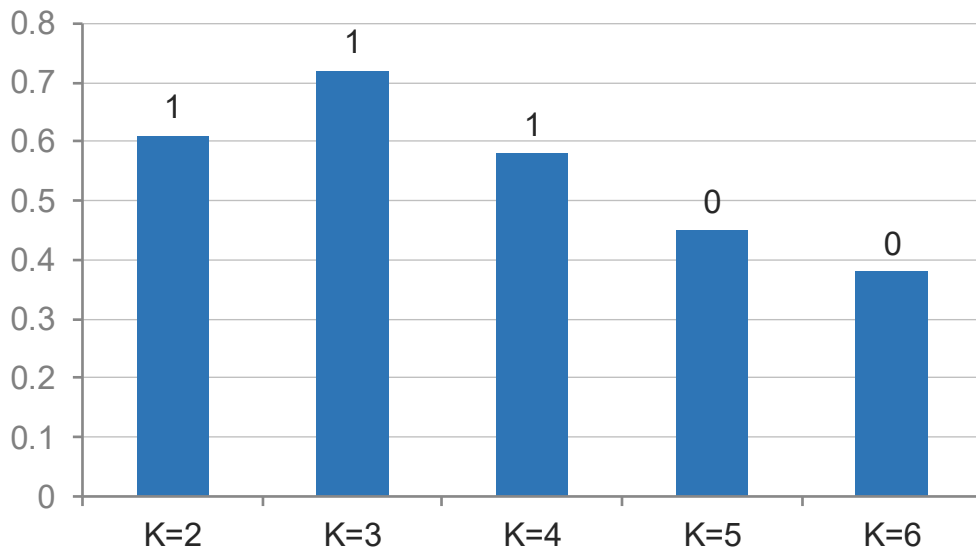
Grid Search: exhaustive but expensive.
Random Search: surprisingly effective.
Bayesian Optimization: smart exploration.
Hyperband: early-stopping bad configs.

Key Hyperparameters

Learning rate, batch size, number of layers, dropout rate, weight decay, optimizer choice, number of epochs.

Silhouette Plots: Bar Charts for Cluster Quality

Knife-shaped bars and the elbow method as complementary visual diagnostics.



Average Silhouette Score by K

What Is Silhouette Score?

Measures how similar a point is to its own cluster vs. neighboring clusters. Range: -1 to +1. Higher = better-defined clusters.

Reading the Plot

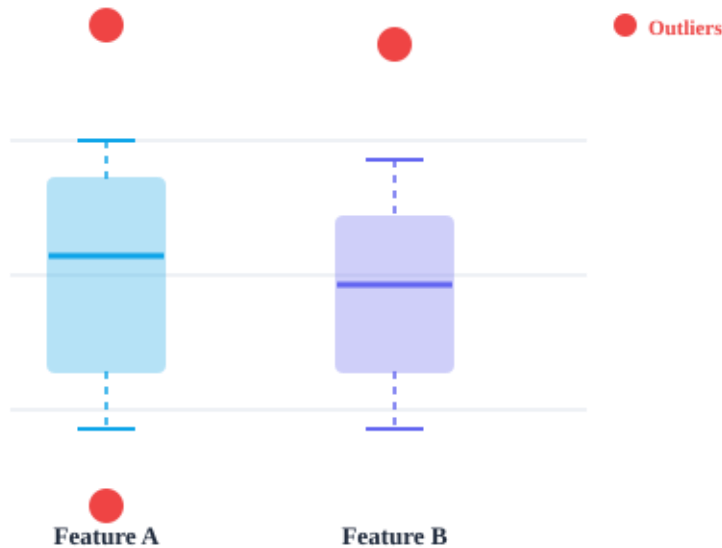
K=3 has the highest score (0.72) → optimal cluster count. The "knife shapes" show each cluster's cohesion and separation.

Complement With

Elbow Method (inertia plot), Gap Statistic, Davies-Bouldin Index. Use multiple methods for confidence.

Box Plots & Scatter for Outlier Detection

Visual encodings for anomalies: position, color, isolation, and annotation.



Spotting Anomalies

Red dots = values beyond $1.5 \times \text{IQR}$ from the box edges. These could be data errors, rare events, or genuinely interesting observations (fraud, defects).

Detection Methods

IQR (Box Plot): simple, visual.
Isolation Forest: tree-based, scalable.
LOF (Local Outlier Factor): density-based.
DBSCAN: clustering-based detection.

Decision

Don't blindly remove outliers! Investigate first. Outliers in fraud detection ARE the signal, not the noise.

Why Visual Explainability Matters

XAI as a visualization problem: making black-box decisions visible.



"Why was my loan denied?"



"Computer says no."

Trust

Users need to understand why decisions are made and feel confident in outcomes.

Compliance

GDPR Article 22 grants EU citizens a legal right to explanation for automated decisions.

Debugging

Engineers need visibility to detect bias, fix errors, and improve models iteratively.

Two Scales of Visual Explanation: Local vs. Global

Choosing the right visual for individual vs. population-level insights.

GLOBAL INTERPRETABILITY

"How does the model work overall?"

- Feature importance rankings
- Partial dependence plots
- Decision tree structure
- Model coefficients/weights

Example: "Income is the most important feature, contributing 35% to predictions across all applicants."

LOCAL INTERPRETABILITY

"Why did it make THIS decision?"

- SHAP values for one prediction
- LIME explanations
- Counterfactual reasoning
- Attention weights on input

Example: "John's loan was denied because his debt-to-income ratio (0.65) exceeded the threshold of 0.43."

SHAP Force Plots: Waterfall Diagrams for Attribution

Stacked-bar and force-plot designs showing how features push predictions.

SHAP Force Plot: Loan Approval Prediction



Shapley Values (Game Theory)

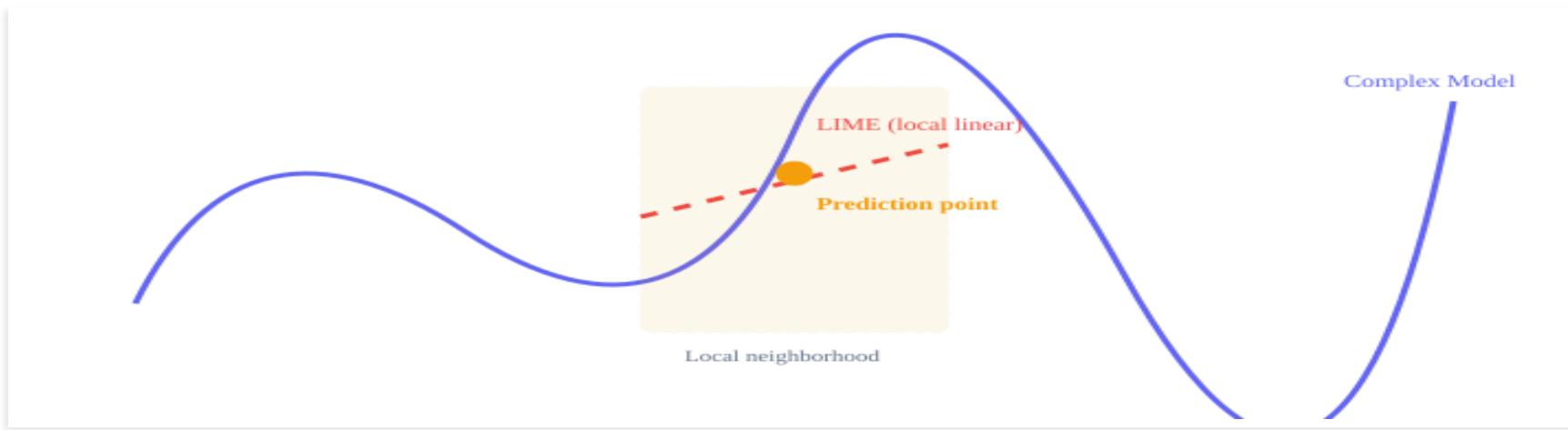
From cooperative game theory: "What is each player's fair contribution to the total payout?" Applied to ML: each feature is a "player" and the prediction is the "payout." SHAP computes each feature's marginal contribution.

SHAP Plot Types

Force Plot: single prediction explanation.
Summary Plot (Beeswarm): feature importance across all predictions.
Dependence Plot: how one feature affects predictions.
Waterfall: sequential feature contributions.

LIME: Visualizing Local Linear Approximations

Overlay diagrams showing simple models fitted within complex boundaries.



How LIME Works

1. Select a prediction to explain.
2. Generate perturbed samples nearby.
3. Fit a simple (linear) model to these samples.
4. The simple model's coefficients = explanation.

Key Advantage

Model-agnostic: works with ANY classifier (neural nets, random forests, SVMs, ensembles). You don't need to understand the model's internals — LIME treats it as a black box.

Saliency Maps: Heatmap Overlays on Input Images

Comparing correct vs. biased visual attention as a diagnostic technique.

CORRECT: Ears & Face



The model highlights the dog's ears, nose, and face shape. These are the biologically relevant features for "dog" classification.

BIASED: Background Grass



The model highlights grass and background. It learned a shortcut: dogs appear in grass, so grass = dog. This is a spurious correlation that fails in new environments.

Methods: Vanilla Gradients, Grad-CAM, Integrated Gradients, SmoothGrad

Text Highlighting as Visual Attribution

Color-coded inline highlighting to reveal which tokens triggered a decision.

Spam Filter: Which words triggered the classification?

Congratulations!

You've

WON

a

FREE

iPhone!

Click **HERE** to claim your prize!

Red highlighted = high attribution to SPAM classification

How Text Attribution Works

Integrated Gradients: compute gradient of prediction w.r.t. each input token embedding, integrate from baseline.

Attention Weights: which tokens the model attended to most.

Occlusion: mask each word and measure prediction change.

Applications

Spam detection explanation.

Sentiment analysis debugging.

Toxicity filter auditing.

Medical NLP: which words in a clinical note triggered a diagnosis code.

Legal AI: which clauses triggered risk flags.

Counterfactual Panels: Side-by-Side Comparison

Before/after visual layouts answering what would need to change.

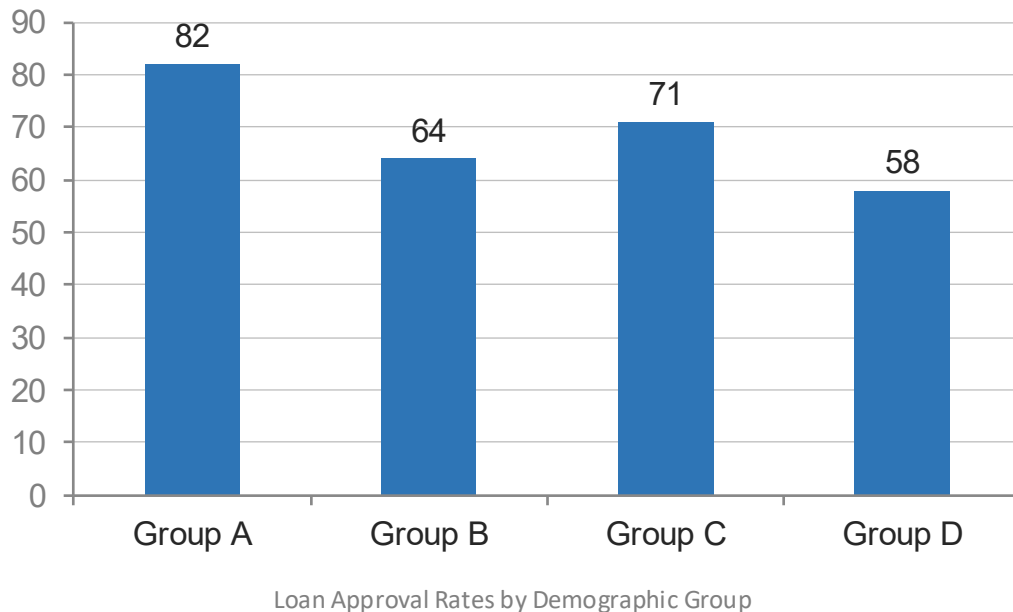
CURRENT: DENIED ✕		COUNTERFACTUAL: APPROVED ✓
Income: \$45,000 Credit Score: 620 Debt-to-Income: 0.55 Employment: 2 years Prediction: Denied (0.32 approval)	→ IF	Income: \$50,000 (+\$5K) Credit Score: 620 Debt-to-Income: 0.43 (-0.12) Employment: 2 years Prediction: Approved (0.71 approval)

The Power of Counterfactuals

Instead of "why was I denied?", counterfactuals answer "what would I need to change to get approved?" — actionable, concrete, human-friendly explanations. Tools: DiCE (Microsoft), Alibi Explain, Carla.

Visualizing Algorithmic Bias: Grouped Bars & Gaps

Design choices for surfacing demographic disparities in model performance.



Disparate Impact

A 24-point gap between Group A (82%) and Group D (58%) signals potential bias. The 80% rule: if the lowest group's rate < 80% of the highest, further investigation is required.

Fairness Metrics

Demographic Parity: equal rates across groups.

Equalized Odds: equal TPR and FPR.

Calibration: equal precision across groups.

Individual Fairness: similar people → similar outcomes.

Tools

IBM AI Fairness 360, Google What-If Tool, Microsoft Fairlearn, Aequitas (UChicago).

Dashboard Design: Weights & Biases (W&B)

Metric cards, sparklines, and run-comparison as visualization patterns.



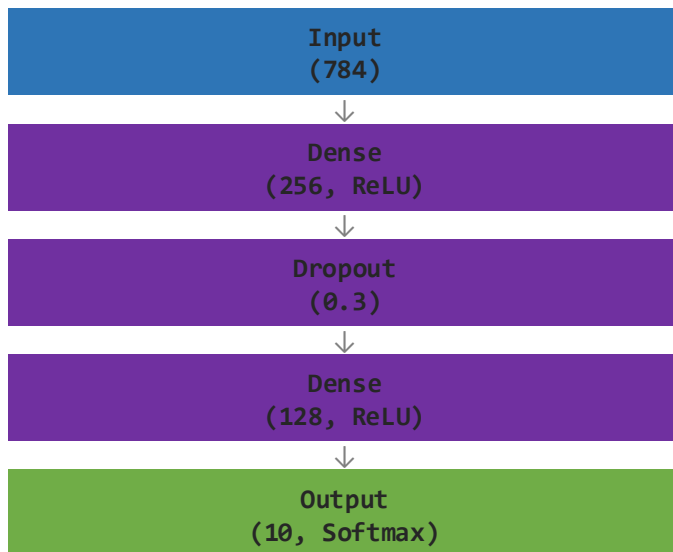
Key Features

Real-time experiment tracking, hyperparameter sweeps, model versioning, team collaboration, artifact management, and reporting.

Graph Visualization in TensorBoard

Node-link diagrams for neural architectures — a direct course application.

Computational Graph View



TensorBoard Views

Scalars: loss, accuracy over time.
Graphs: model architecture visualization.
Histograms: weight distributions.
Images: sample predictions.
Projector: embedding visualization.

Getting Started

```
tensorboard --logdir=./logs  
In code:  
tf.summary.scalar("loss", loss)  
tb_callback = TensorBoard(log_dir)
```

Pro Tip

Use `tf.profiler` for GPU utilization and bottleneck detection during training.

Interactive Dashboards: Streamlit & Gradio

Building visual interfaces for non-technical users — the last mile of viz.

STREAMLIT



Loan Prediction App

Income: [=====|====] \$52,000

Credit Score: [=====|==] 720

Predict



Approved (87% confidence)

~10 lines of Python to build this

GRADIO

Image Classifier



Drag & drop image here

Dog: 92% | Cat: 6% | Bird: 2%

Auto-generates shareable link



The Future of Visual Analytics for AI

"Natural Language as the UI for Data Science"

AIGC Everywhere

Every creative workflow will have AI co-pilots. Content creation costs approach zero.

Transparent Models

XAI becomes standard. No black-box decisions in healthcare, finance, or justice.

Visual Intelligence

Visualization bridges the gap between model complexity and human understanding.